npn junction

$i_C = I_S e^{V_{BE}/V_T}$

↳ saturation current

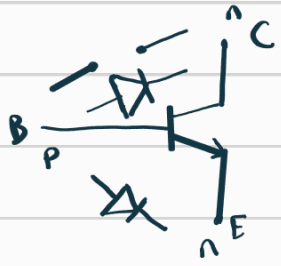


$$i_B = \frac{i_C}{\beta} \qquad i_E = \frac{\beta+1}{\beta} i_C = \frac{1}{\alpha} i_C = \beta+1\, i_B$$

$$i_C = \alpha\, i_E$$

	Active	Sat	Cutoff
EBJ	Forward bias	Forward bias	Reverse bias
CBJ	Reverse bias	Forward bias	Reverse bias

In npn

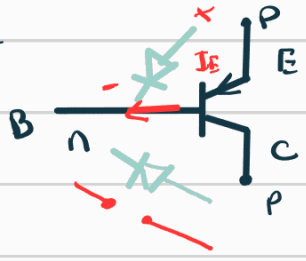


Active: $V_{BE} \geq 0,7$, $V_{BC} \leq 0,4V$

Cutoff: $V_{BE} < 0,7$, $V_{BC} \leq 0,4V$

Saturation: $V_{BE} \geq 0,7$, $V_{BC} > 0,4V$

In pnp

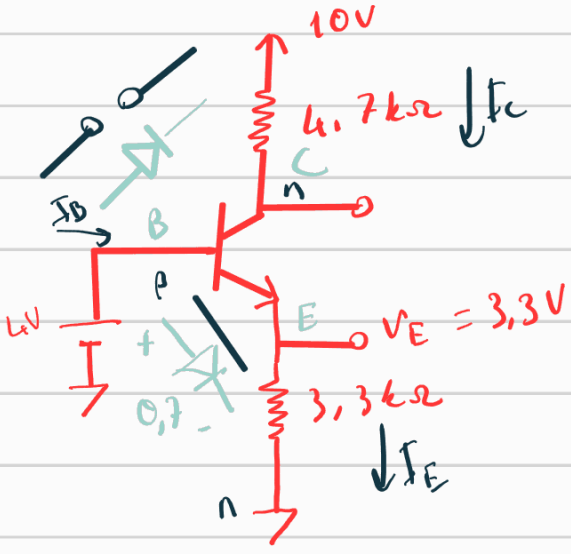


Active: $V_{EB} \geq 0,7$, $V_{CB} \leq 0,4V$

Cutoff: $V_{EB} < 0,7$, $V_{CB} \leq 0,4V$

Saturation: $V_{EB} > 0,7V$, $V_{CB} > 0,4V$

Ex1 Analyze the circuit below calculate all node voltages and branch currents. Assume that β is specific to be 100. $V_{BE} = 0,7\ V$



npn

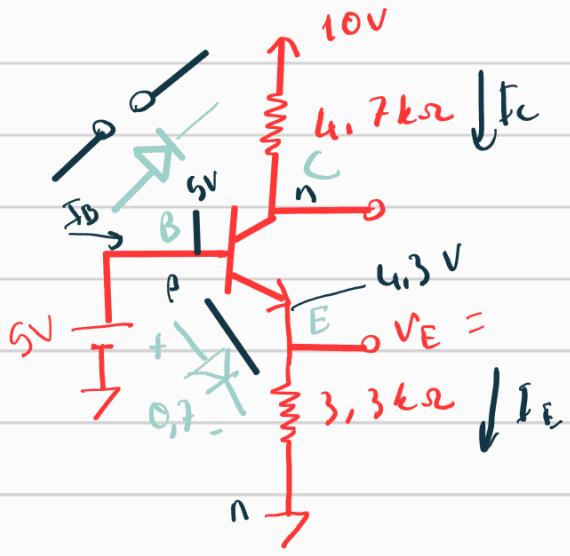
Assume that transistor operate in active mode.

$$I_E = \frac{3,3V}{3,3k} = 1mA$$

$$I_E = \frac{1}{\alpha} I_C$$

$$I_C = \alpha I_E = \frac{\beta}{\beta+1} I_E = \frac{100}{101} 1 = 0,99mA$$

$$I_E = I_B + I_C \qquad I_B = I_E - I_C = 1mA - 0,99mA = 10\mu A$$



npn

$V_{CE} = 0,2V$

Assume in Activemode

$$I_E = \frac{4,3}{3,3k} = 1,30mA$$

$$I_C = \frac{\beta}{\beta+1} \cdot 1,30 = 1,287mA$$

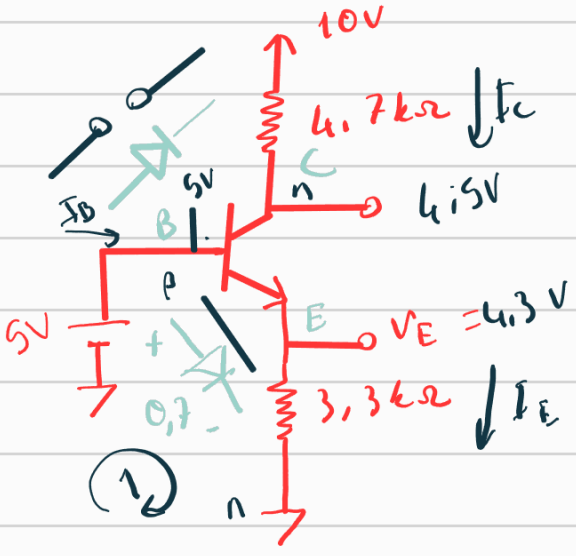
$$\Rightarrow V_C = 10 - (4,7 * 1,287)$$

$$= 3,95$$

$$V_{BC} = 1,05 \overset{?}{\geq} 0,4V$$

X

Not in activemode



npn

I_n sat = ?

V_{CE} sat = 0,2V

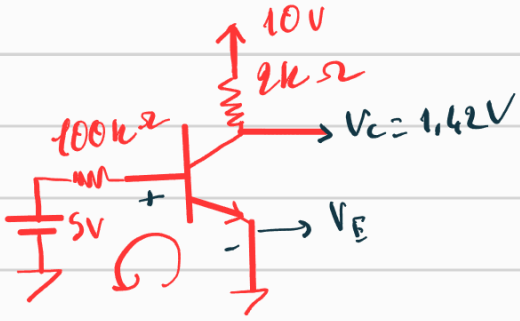
$$I_E = 1,30mA$$

$$I_C = \frac{10-4,5}{4,7k} = 1,17\,mA$$

$$I_B = 0,13mA \;//$$

$$\beta sat = \frac{I_C}{I_B} = \frac{1,17}{0,13} = 9 \;//$$

Ex 2 Calculate all node voltages and branch currents for the circuit below $V_{BE} = 0,7V$, $\beta = 99$



$$-5 + 100k I_B + 0,7 = 0$$

$$I_B = \frac{4,3}{100k} = 43\mu A$$

$$I_C = \beta \cdot I_B = 4,29mA$$

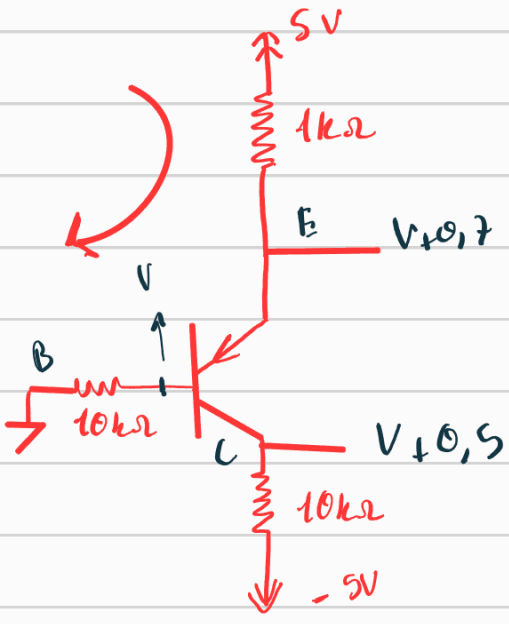
$V_C = 10 - (4{,}28\,mA \cdot 2k\Omega) = 1{,}42V$ //

$V_B = 5 - 100 \cdot 43\,mA$
$= 5 - 100 \cdot 43\,mA$
$= 5 - 43\,A$
$= 0{,}7\,V$

$V_{BC} \leq 0{,}4$ ✓

Ex 3 Calculate all branch currents and node voltages

$\beta = 30$



$V_{EC\,sat} = 0{,}2V$

Assume inactive mode

$-5 + 31 I_B + 0{,}7 + 10k I_B = 0$

$41k I_B = 4{,}3V$

$I_B = 0{,}11mA,\ 110\mu A$

$I_C = 3{,}3mA$

$I_E \approx 3{,}4mA$

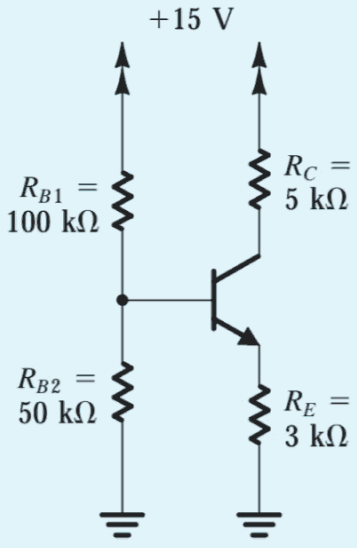
$V_C = 28\,V$ not possible

Assume sat

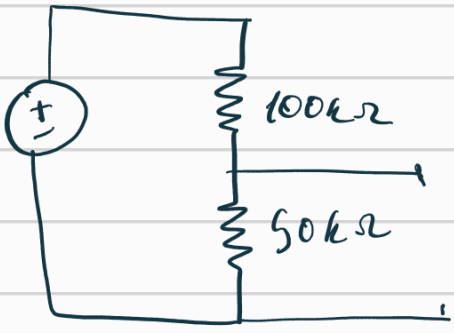
☐

Example 6.10

We want to analyze the circuit of Fig. 6.28(a) to determine the voltages at all nodes and the currents through all branches. Assume $\beta = 100$.



(a)



$V_{oc} = \frac{50}{150} \cdot 15$

$= \frac{50}{10} = 5V$



$\frac{15}{100k} = 0,15mA$

$V = I \cdot R$

$R_{th} = \frac{5V}{0,15mA} = 33,3k\Omega$

